DATA 602: Final Project Proposal

Predicting Population Health Data to Predict Risk Factors for Disease

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1. Business Challenge

Diabetes and heart disease remain leading causes of death and healthcare expenditure. Proactive evaluation of disease risk through community and patient health metrics enables timely interventions and preventative care. This project uses health, lifestyle, and demographic data from national sources such as CDC and HealthData.gov to develop a classification-based machine learning model that predicts disease likelihood. Insights will help policymakers and healthcare professionals identify high-risk groups and design targeted wellness programs.

2. Data Sources

• BRFSS (CDC): self-reported health issues, demographics, lifestyle factors.
• HealthData.gov: public health statistics on chronic illnesses by county.
• NHANES: physiological and biometric measures to improve model accuracy.

3. Methodological Framework

Supervised classifiers to predict disease incidence: Logistic Regression (interpretable baseline), Random Forest (non-linear relationships; feature importance), Gradient Boosting (iterative gains). Hyperparameter tuning via GridSearchCV or Optuna. Metrics: Accuracy, Precision, Recall, F1-score, ROC-AUC.

4. Experimental Methodology (CRISP-DM)

Business Understanding: Defining goals and success metrics. Data Understanding: explore BMI, smoking, activity, blood pressure indicators. Data Preparation: impute missing values, encode categoricals, and normalize numerics. Modeling: train and optimize the three classifiers. Evaluation: validate on held-out test data. Deployment: present insights via Power BI dashboards and summary reports.

5. Assumptions and Restrictions

Datasets approximate population coverage though some illness classes may be underrepresented. All data are anonymized and comply with HIPAA and open data policies. Imputation/normalization addresses missing values and outliers. Outcomes are correlational, not diagnostic.

6. Adjusted Work / Novelty

Integrates multiple national datasets and applies ensemble learning to improve forecasting accuracy. Novelty: a unified pipeline combining demographic, lifestyle, and physiological indicators to identify significant health risk factors.

7. Requested Deviations

None. The project adheres to DS602 requirements: originality, model experimentation, and a final technical presentation.

8. Expected Outcomes

A predictive model for disease risk probabilities; charts of key predictors (BMI, smoking, exercise); a Power BI dashboard for geographic risk; and actionable insights for public health policy and wellness interventions.